# Angry Balloon

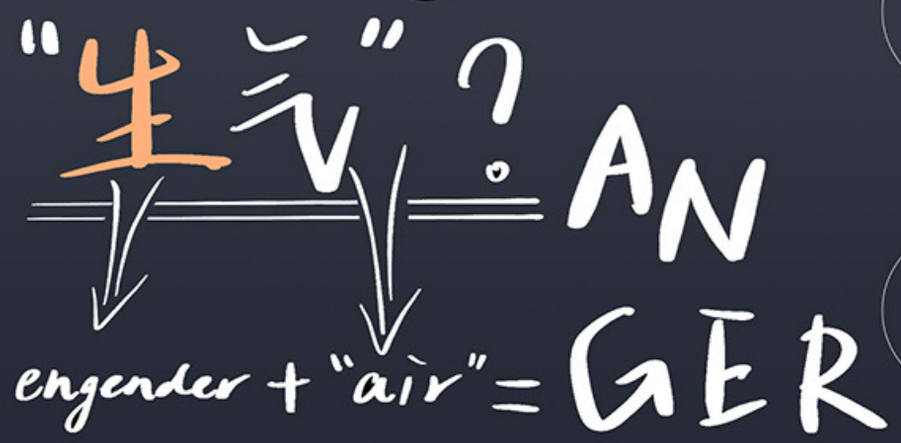
*KEY WORDS: Interactive installation / Negative Emotions / brainwave*

*An interactive installation*
*According to the emotional problems which easy come easy go, I attempt to digitalize negative thoughts of my subjects.*
*Simultaneously, I try to distract subjects to make them forget bad mood by showing visualization and startling outcome.*

## INITIAL IDEA

This project is inspired by a brainscience project. When I was working on this project, I was in a bad mood as I had a fight with my good friend. Hence, I start wondering that whether I can eliminate negative emotion by visualizing brainwaves, since brainwaves are detectable and distinguishable?

In Chinese, **ANGER** is represented by word **SHENG QI**. **QI** means **AIR**, how do we spotit and how do we get rid of it?




## RESEARCH

As we all know, we are irritable during puberty because of unstable emotion state. So I interview six schoolmates, trying to figure out the cause of anger, the duration of anger and how this emotion ends.

Sombra / 18
Freshman

"I am always angry on stuff on the Internet, but just last for a moment, when the next video is hilarious"

Liao / 17
12th Grader

"Mostly because of parents or friends, and I always forget it very soon, there's no necessity to argue"

Qu / 17
12th Grader

"I always angery with people around me and very rude to me, but when they trying to be nice, I am fine with that"

Tania / 17
12th Grader

"when I realized that i am not being respected, and i remember it in the long haul"

Aphra / 17
12th Grader

"lost things always make me angry, and it last until i find it back"

Emma / 18
12th Grader

"when others break the promise and make excuses. But as long as they try to please me afterwards I will calm down soon."

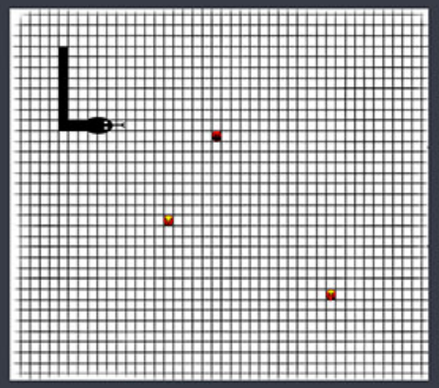
## CONCLUSION

01 **Most of the anger of teenagers is triggered by trivial matters**

02 **The mood easy come easy go and leave little effect, distraction is the key to anger**

# SOLUTIONS

## PLAN A (Gluttous Snake Eating Away)



Eating is a prevalent method to relieve ourselves when we are in a mood. Thus, I wish to visualize negative energy in daily lives and consume those annoying stuff through brainwaves and head movement.

## PLAN B (Transforming the Mood & Eliminate) - Final

Anger = Sheng Qi (in Chinese)
Balloon = Qi Qiu

What if they were the same Qi?

I would like to design a hat that, when worn, can identify our negative emotions through its built-in detectors. Then, the "gas" of these negative emotions will be used as filling for balloons to visualize our negative thoughts.

# TEST

I managed to export different brainwave patterns of various emotion by brainwave detector:

## / JOYFUL /

Theta waves have a frequency between 4-7Hz. They are regular to teenagers in the age range of 10 to 17.

## / FOCUSED/

Gamma waves have a frequency between 30-47Hz, the highest frequency among brainwaves. When focused on a specific thing or staying alert, we are easy to have gamma waves.

## / IRRITATED /

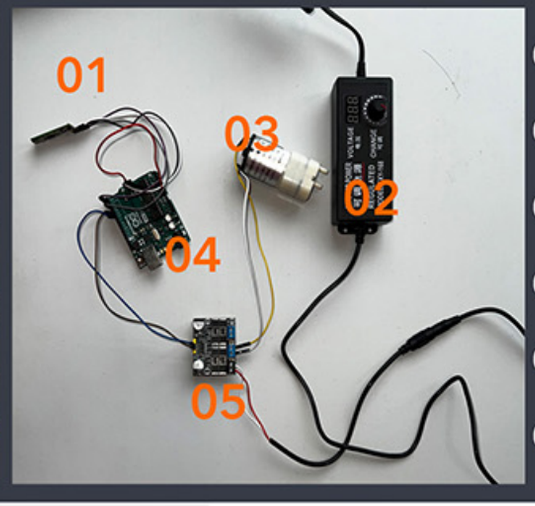
Having a frequency between 13-30Hz, beta waves play the main role in logical thinking and are concentrated in frotal lobe. Our brains produce them when we are sober, tense and excited.

## SUMMARY

People when experiencing the same emotions, but there are still some differences. I will attempt to visually represent negative emotions through data transformation.

# INSTALLATION ASSEMBLING & PROTOTYPE

## Arduino and EEG sensor connection




01 Bluetooth Module

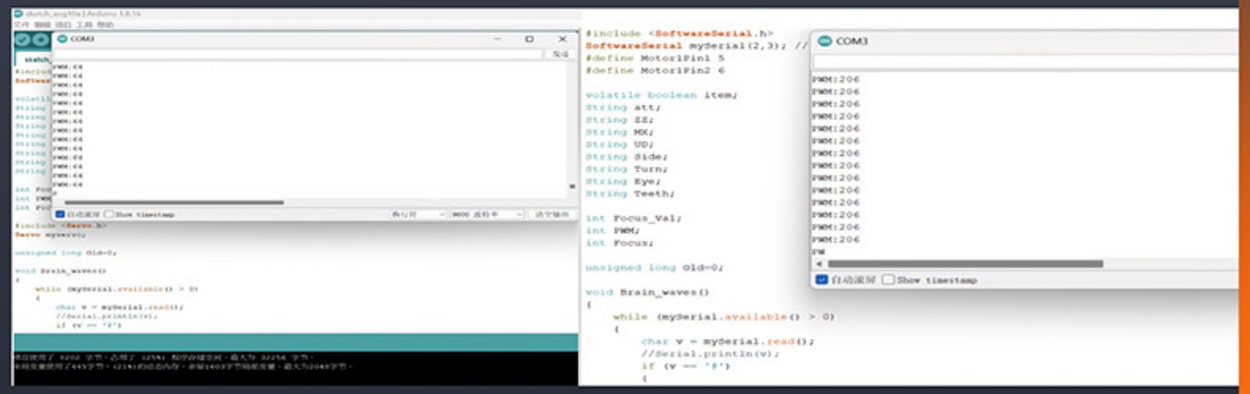
02 Source

03 Air Pump

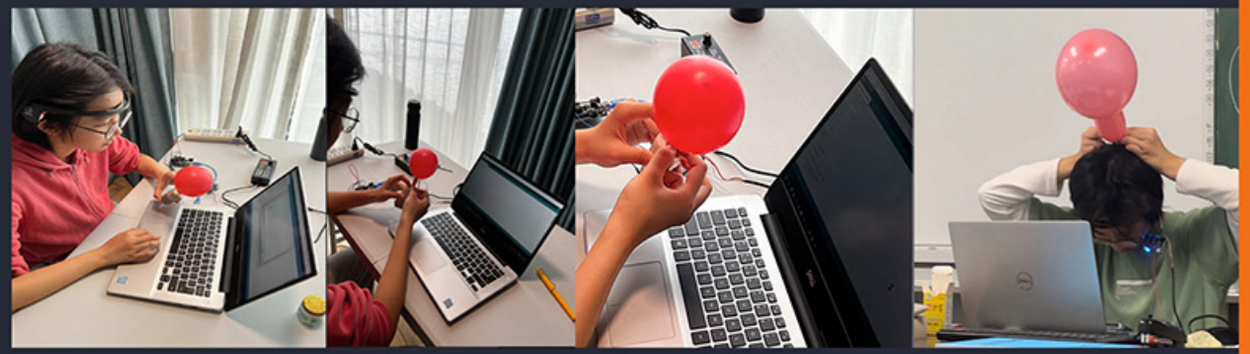
04 Arduino

05 DC Motor Drive

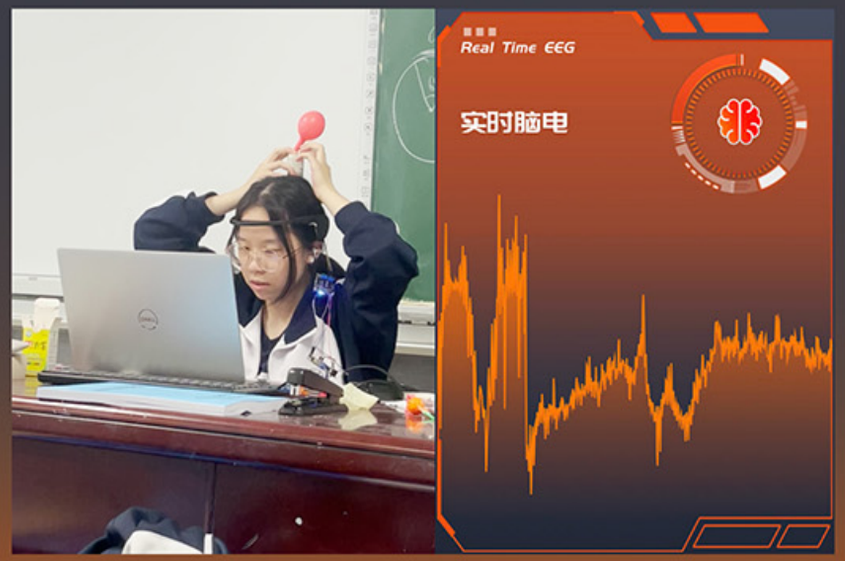
06 EEG Sensor




## Linkage between air pump and EEG data



# BEFORE




# AFTER

# FINAL

Jingshan He

Year of Creation: 2023
Medium: Adobe Software / Hand Drawing
Individual Design Project